Mohini Deshpande

AI Engineer building agentic systems, RAG pipelines, and production ML — from architecture through daily use.

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About

AI and software engineer with an MSc in Advanced Computer Science: AI and about two years of experience building production AI systems in Python: LLM agents, retrieval-augmented generation, and the backend services around them. I also build and evaluate predictive models, and I take projects end to end, from requirements gathering through daily production use.

Expertise

Languages & Frontend
Python, JavaScript/TypeScript, SQL, MySQL, React, HTML, CSS
Backend & Data
FastAPI, Node.js, REST APIs, PostgreSQL, Redis, Celery, MongoDB, pandas, vector databases, Qdrant, pgvector
AI, LLM & Agentic Systems
Google ADK, Gemini API, Llama API, LangChain, Hugging Face, Model Context Protocol, retrieval-augmented generation, prompt engineering, multi-agent orchestration
Machine Learning & NLP
PyTorch, TensorFlow, scikit-learn, XGBoost, neural networks, transformers, BERT, FinBERT, NLP, model evaluation, cross-validation
Cloud, DevOps & Tools
AWS, GCP, Azure ML, Docker, Kubernetes, CI/CD, GitHub Actions, Git, Claude Code, GitHub Copilot
Automation & Data
n8n, Make, Zapier, UiPath, Microsoft Copilot, Excel data cleaning and reconciliation

Experience

Contract Software Engineer — *Ekblink* Jul 2026 – Present

- Building Python backend infrastructure with FastAPI, Redis, Celery, and PostgreSQL for a production AI content-moderation pipeline, with logging and monitoring.

Contract Software Engineer — *Independent Client, India* Dec 2024 – Mar 2025

- Sole engineer of a Python and Gemini API agent that auto-drafts replies to roughly 65–70% of 45–50 weekly client emails and routes the rest to a human.
- Tested with pytest and DeepEval before shipping; in daily production use since delivery.

AI Trainer — *Outlier, Remote* Oct 2024 – Jul 2025

- Designed evaluation frameworks across 150+ AI model responses, lifting output credibility by 20%, and turned findings into clear recommendations for stakeholders.

Machine Learning Intern — *SmartKnower, India* Apr – May 2021

- Built Python ETL pipelines and CNN/LSTM neural network models, lifting data quality by 25% and improving business metric projections by 12%.

Projects

Bank Marketing Prediction — *leakage-free model comparison* github.com/MohiniDeshpande/bank-marketing-prediction

- Compared five classifiers with cross-validated pipelines; XGBoost reached 0.76 held-out ROC-AUC, and the top 20% of clients by score capture 57% of subscribers, 2.8 times random calling.

FixIt AI — *autonomous agentic assistant* github.com/MohiniDeshpande/ai-handyman-backend

- Agentic assistant built with Google's Agent Development Kit and FastAPI, using tool calling and a RAG pipeline over Qdrant and pgvector.

PokéMarket — *AI market intelligence platform* github.com/MohiniDeshpande/pokemarket-ai

- Agentic chatbot with a React frontend, using MongoDB MCP and the Gemini API for live market data, containerised with Docker and deployed with CI/CD.

Stock Trend Prediction with News Sentiment — *FinBERT and Random Forest*

- Combined FinBERT news sentiment with technical indicators such as RSI and moving averages to predict price trends with a Random Forest, reaching 67% accuracy.

Education

MSc Advanced Computer Science: AI
University of Manchester, UK — 2023–2024
Dissertation: Animation of Avatar Using Human Motion Capture Data — motion graphs, point clouds, OpenGL visualisation
Coursework: Reinforcement Learning, Deep Learning, NLP, Computer Vision, Data Engineering

BSc Computer Science
University of Mumbai, India — 2019–2022
Dissertation: Restaurant Recommendation System Using Machine Learning and Flask

Certifications

Google Cloud Platform Fundamentals — Coursera

ML Pipelines with Azure ML Studio — Coursera

UiPath Workflow Automation Certification

Ethics of AI — University of Helsinki, in progress